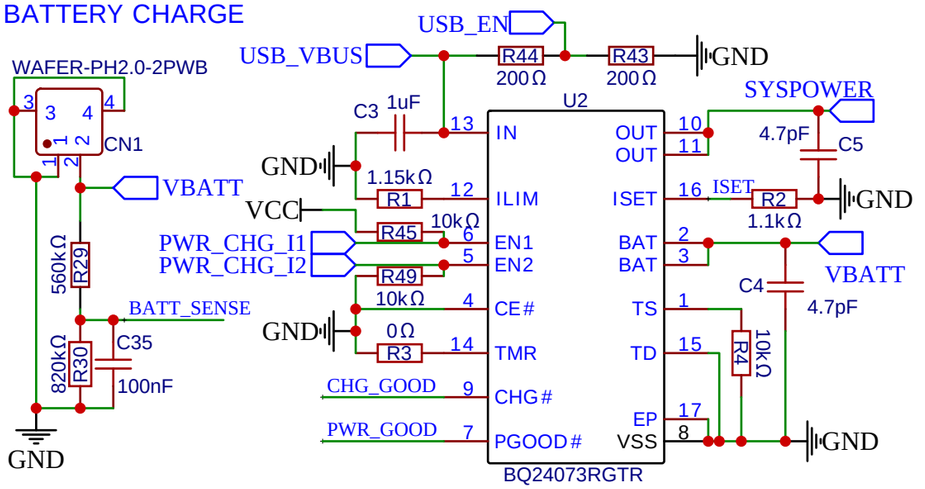
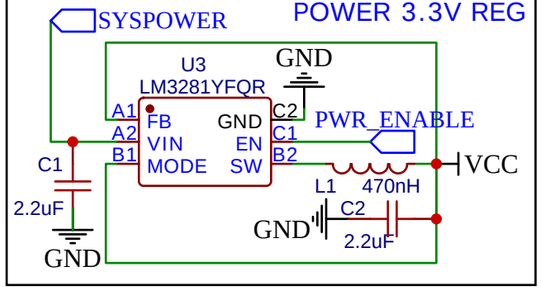
POWER SYSTEMS

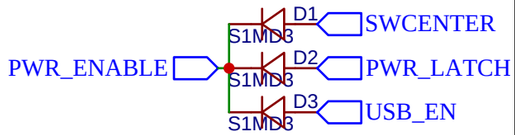
BATTERY CHARGE



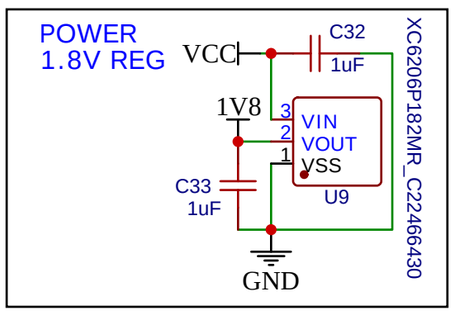
POWER 3.3V REG



POWER TURN ON / LATCH

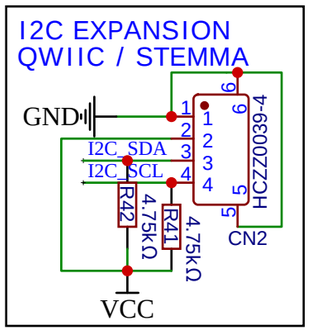


POWER 1.8V REG

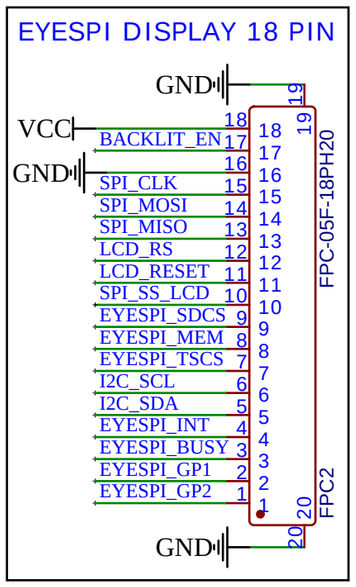


INTERFACE & EXPANSION PORTS

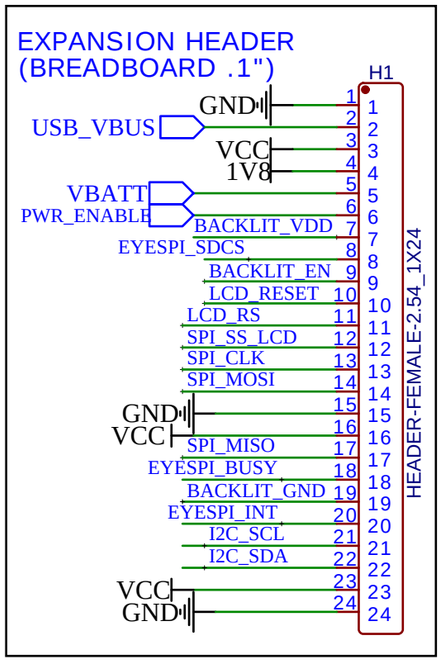
I2C EXPANSION QWIIC / STEMMA



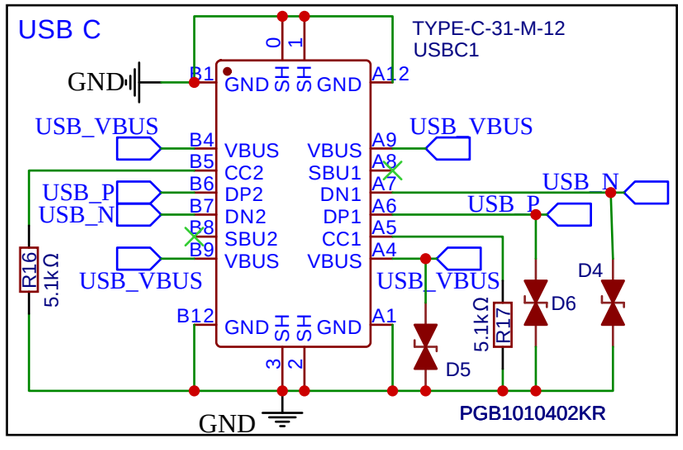
EYESPI DISPLAY 18 PIN



EXPANSION HEADER (BREADBOARD .1")

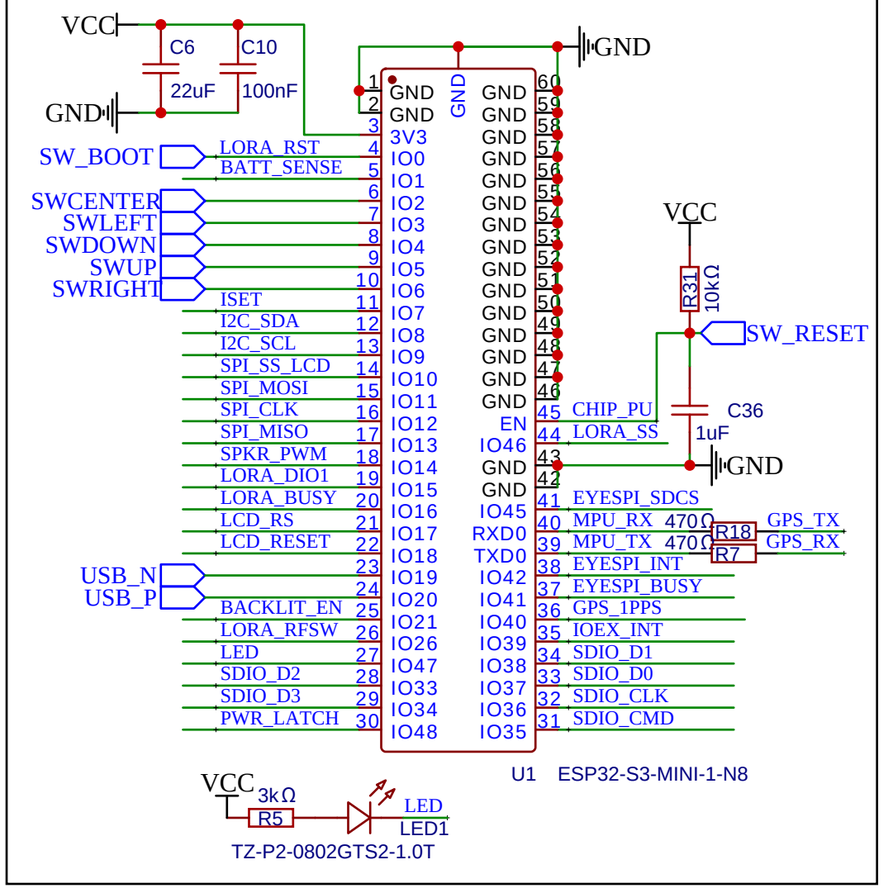


USB C

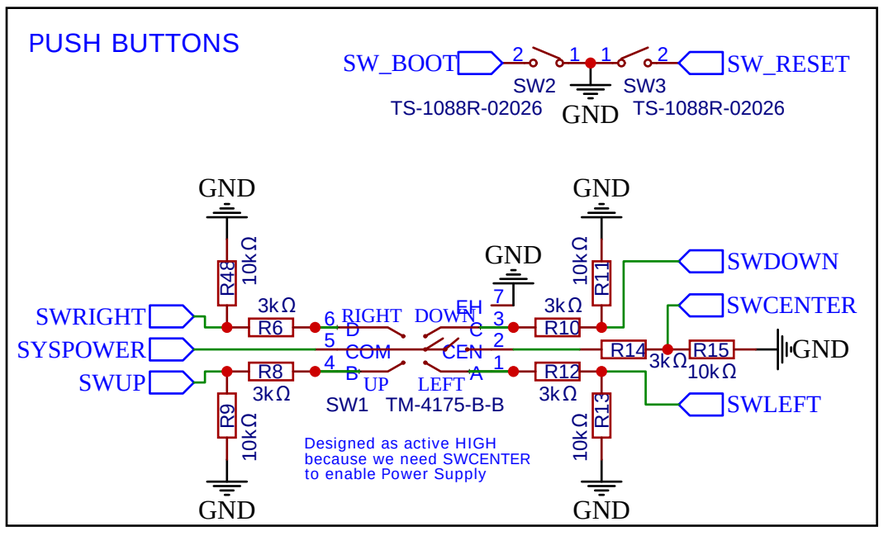


PROCESSOR & BUTTONS

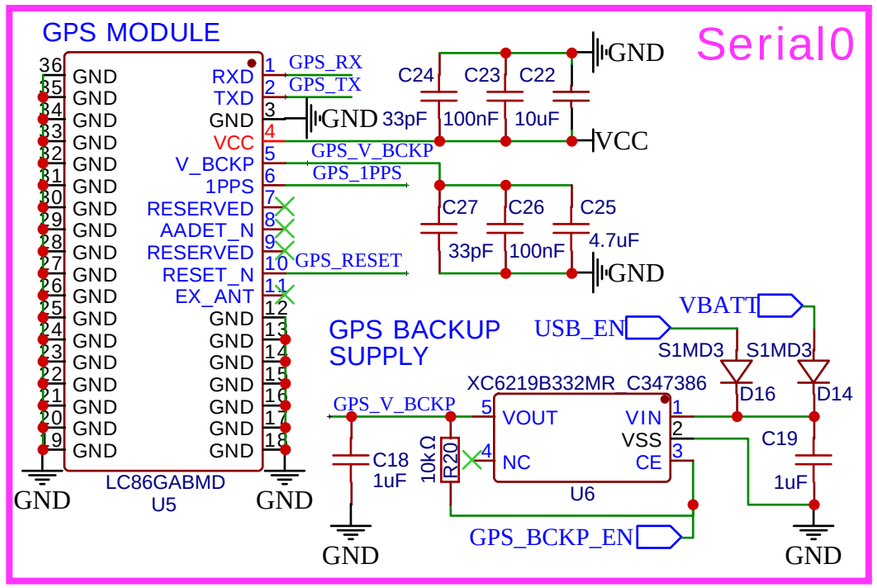
MAIN PROCESSOR ESP32-S3-FN8

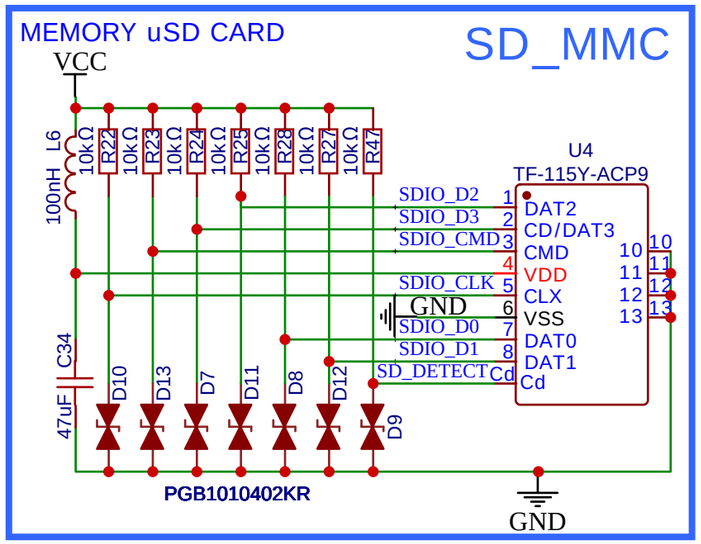


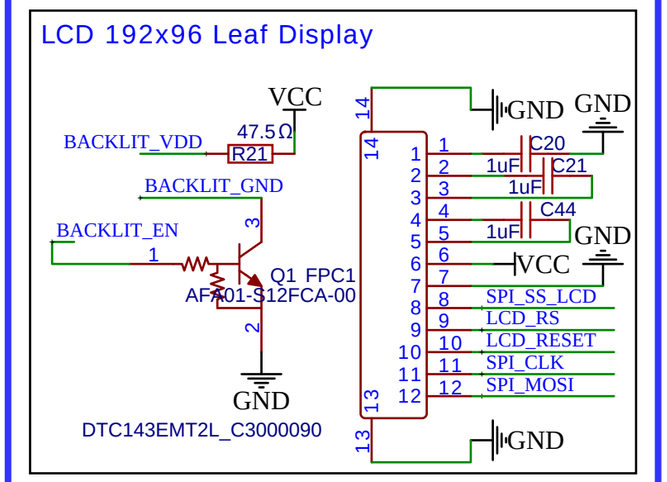
PUSH BUTTONS

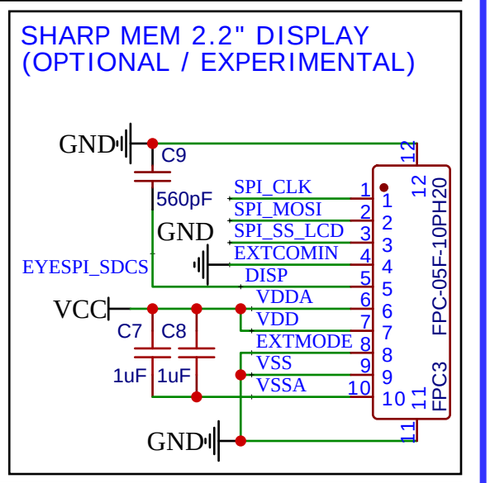


TITLE: Main System		REV: 3.2.8
EasyEDA	Company: Leaf	Sheet: 1/3
	Date: 2026-10-20 Drawn By: oxothnk	

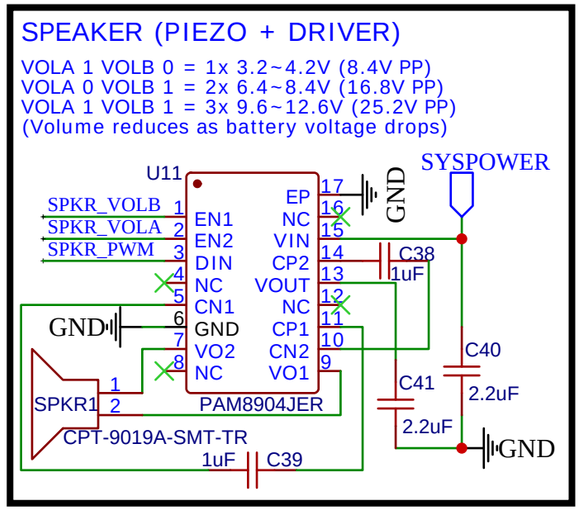


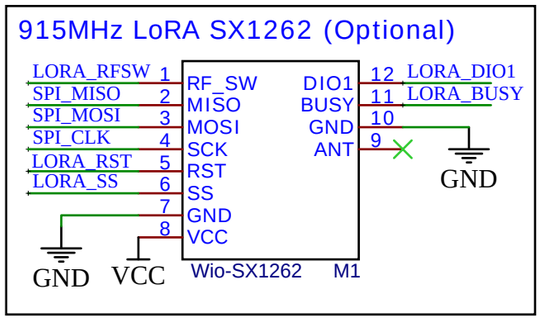




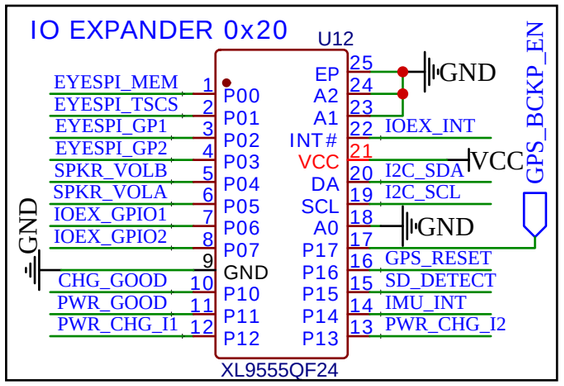


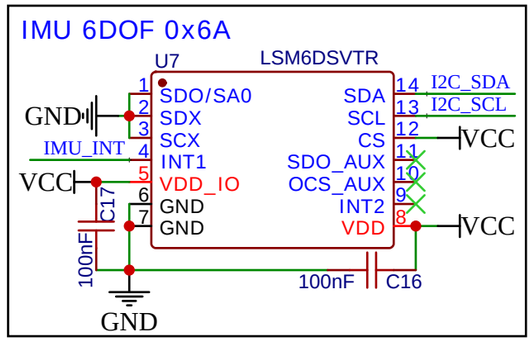
SPI DEVICES

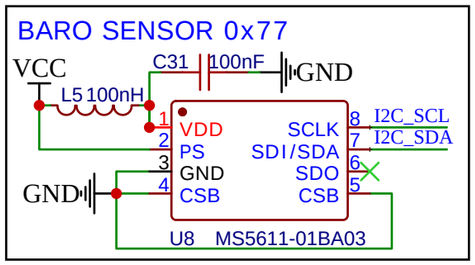


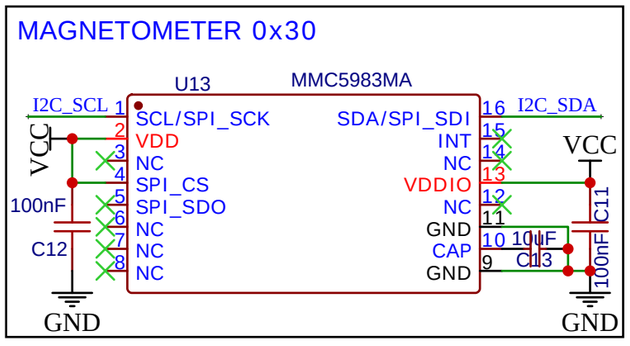


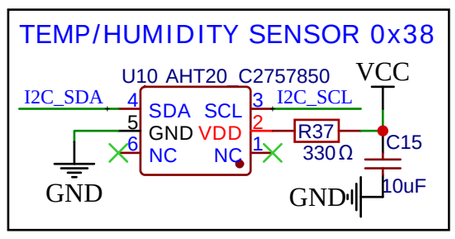
I2C DEVICES











TITLE: Peripherals	REV: 3.2.8
EasyEDA	Company: Leaf / Sheet: 2/3
	Date: 2026-10-20 Drawn By: oxothnk

Three ways to "turn on" the system:
1. USB-C plugged in
2. Hold center button
3. [once it boots] the processor can latch-on the power supply

BUTTONS
CENTER
LEFT
DOWN
UP
RIGHT
RESET
BOOT

EXPANSION CONNECTORS
4Pin I2C STEMMA/QWIIK
18Pin EYESPI
24Pin Breadboard 0.1"

MAIN 3.3V REGULATOR
ON/OFF PWR_EN

PROCESSOR ESP32 S3 FN8

USB C

CHARGE MANAGEMENT

BATTERY

SERIAL 0

GPS

I2C BUS

IMU 6 DOF ACCEL GYRO

MAG/COMPASS

BAROMETRIC PRESSURE SENSOR

TEMP & HUMIDITY SENSOR

IO EXPANDER

SD_MMC BUS (4 WIRE)

SPI BUS

LCD DISPLAY

LoRa DATA RADIO (OPTIONAL)

SD CARD

SPEAKER DRIVER

SPEAKER

The speaker is powered directly from the charger output to gain maximum volume: 4.4V if charging, otherwise direct battery voltage (3.3~4.2V). This is louder than just using the 3.3V regulator at all times. Note the volume will decrease ~25% as the battery discharges (4.2 down to 3.3V)

Leaf is safe up to 9V input on the USB-C port.

This allows use of "5V" solar panels, which can often output anywhere from 4V to 8V.

TITLE: Block Diagram	REV: 3.2.8
EasyEDA	Company: Leaf / Sheet: 3/3
	Date: 2026-10-20 Drawn By: oxothnk